American Airlines Flight 96

Flight model: Douglas DC 10
Date: June 12, 1972

From: Los Angeles, US
To: New York, US

Problem

The scheduled flight is going to New York with a stop over at Detroit and Buffalo both in the US. There are 2 active pilots and another pilot on standby. It is worth nothing that the DC-10 was just introduced in the same period therefore the pilots have less than 70 hours of flying in the same. The plane has already covered the distance from LA (Los Angeles) to Detroit. The plane takes off from Detroit, the plane was still climbing. Suddenly the passengers hear a loud bang followed by sever turbulence. The plane also started to roll toward the left. The throttles that control the jet have become idle. The pilots losses control over the jet. The plane started to slow down. The rudder that controls the plane yaw motion is jammed right. The flight attendants found that the floor near the lavatory at the black of the plane had a big hole in it . The attendants were able the see the ground below through the hole. The Douglas DC 10 has 3 engines 2 on each side of the wings and one on the rudder. The engine located on the rudder stops responding which means they have the engine working but were unable to take of the same. An emergency was immediately declared. The elevators in the tail of the plane were also hard to control, they did respond but, the response time for the same was high. Without the rudder the pilots uses engines to control the plane but the problem with this approach is that the changes to the plane motion to take place would require time. The plane starts to fall very rapidly. The pilots indent to land the plane in the nearest airport. The power is increased to slow the rate of descend. Despite high forward airspeed the plane manages to land in the runway safely.

Investigation

In the investigation, it was found that few of the cargo that the plane was carrying along with the cargo hold door was found far away from the airport. Most of the doors in a plane is opened from inward because the air pressure difference creates a suction and firmly closes the door. In the case of the DC-10 the door was opened outward. The door is closed tightly through latches on the door which on further analysis was found to be not latched properly. It was found that the door could be closed even without the locking pins that were holding the latches in place and not indication was there in place for the crew or the ground staff to notice. The decompression of the plane should not have disturbed the controls of the plane. It was studied that the decompression had tore the floor of the passenger compartment which houses few of the important control cables to control the plane. Later, a similar incident occurs in Turkish Airline 981, where in hole appears in the foot of the passenger cabin leading to decompression and finally crashing into the ground. In the very same incident the rea cargo door was broke open and was found 15 KM off the crash site. Which made the investigators suspect the same cause as that of American Airline flight 96. The reports of the Turkish Airline 981 pointed that the cause of the incident was in fact the same as that of American Airline flight 96, which is the improper locking mechanism of the cargo hold door. But, the loss of control in former was much worse than the later (American Airline flight 96), with all its hydraulics destroyed. It was also found that the cargo door manufacturer had warned the plane manufacturer regarding the potential danger of such a design of the door and also suggested few improvement of the same.

The manufacturer before the American airline flight 96 accident, knew that the pane failed in a decompression test and knew the potential consequences but failed to make any changes to its design.

Changes that were brought after the incident

The report advises the Douglas manufacture to make the changes in the cargo hold door and also make sure without the locking pins in place, the cargo hold door cannot be closed and also suggested placing vent in the passenger floor of the plane so that any decompression would not collapse the floor of the plane. Which until the Turkish Airline 981 accident, was not brought into action. This time an **air-worthiness directive** was issued to the manufacturer which implies that without the recommended change be made the plane are classified as unfit for flight by the FAA. Despite the changes being made and proven to be successful, the manufacturer went bankrupt.

Reference

1. "There's A Hole In A Cabin! — American Airlines Flight 96 — Mayday: Air Disaster `https://www.youtube.com/watch?v=Vz17wYdo5vA`